Pulsar: Exact Tree Speculation with Confidence-Based Scheduling

The Pulsar Project

technical report v0.2 — Stage 6 backend study

October 2026

Abstract

Speculative decoding trades draft and verification work against the target model's autoregressive cost. Pulsar's frozen verifier gives a pointwise exactness guarantee: proposing from q, accepting with $\min(1, p/q)$, and drawing a rejection replacement from $(p-q)_+$ preserves the target next-token law p. This guarantee does not imply a throughput gain. The central empirical hypothesis is conditional and falsifiable: under a measured backend cost profile, a draft-confidence scheduler for exact bounded trees improves held-out tokens/s over a tuning-selected fixed tree only if the paired 95% confidence interval's lower bound is positive. A Stage 6 Hugging Face Transformers CPU integration (distilgpt2 target, tiny-gpt2 draft, temperature-1 softmax, 24-token horizon, three held-out prompts) passed the full-vocabulary verifier identity with maximum absolute error $1.1 \cdot 10^{-19}$. However, the confidence-tree point estimate was +1.66% versus the fixed tree, with 95% CI $[-6.71\%, +10.01\%]$; all confidence bins selected the same fixed action, so the scheduler claim is not established. Tuned $\gamma = 2$ chain speculation reached 15.71 tokens/s and beat the fixed tree (10.98 tokens/s) by 43.00% $[34.87\%, 50.79\%]$, but both were slower than autoregression at 40.62 tokens/s. Earlier 1.39× results are from an analytic cost-model Markov testbed, not measured neural-model acceleration. Pulsar, rotor, and quantum terminology is explanatory framing; those mechanisms are not part of the Stage 6 measured algorithm.

1 Introduction

1.1 The emission bottleneck

Every mainstream LLM is a decoder: each forward pass produces a distribution over V outcomes and then destroys it, emitting a single sample. The engine pays the memory-bound per-token tax because it retains no representation of the *future field* and no principled scheduler for *when* to collapse it. Speculative decoding and its successors (Medusa, EAGLE, Lookahead, Sequoia) draft candidate fields and verify in parallel, but the field is handled ad hoc: fixed speculation lengths, hand-tuned acceptance rules, and tree samplers whose renormalization quietly changes the proposal the verifier must respect.

A pulsar is nature's solution to "emit fast from a huge field": a rotating magnetized star sweeps a beam through the entire sky, observers receive *pulses* whose timing is predictable to extraordinary precision. Three features transfer into an inference engine: (1) a **rotor sweeping a field** — stratified, budgeted coverage of candidates; (2) **pulses, not streams** — commit when local field density (confidence) crosses a threshold, emitting $1..N$ tokens; (3) a **clock independent of content** — per-cycle work is bounded by construction.

1.2 Design stance

The novel, trainable core is deliberately *sub-symbolic and small*: a confidence field (M4c) estimating verifier-acceptance probabilities from draft-side features. Everything correctness-critical

is *frozen*: the rejection-sampling verifier guarantees that no matter how M1, M2, M4a–M4d, M7 misbehave — even adversarially — output remains the base model's distribution. Speed is bought with miscalibration; quality cannot be. This asymmetry is what makes a fast engine shippable, and it is the paper's organizing theorem.

1.3 Contributions

C1 *Superposition fields* (§3): prefix-tree candidate structures with Born-shaped path amplitudes; "quantum" language bounded to diagonal density operators; the branching domain is the load-bearing object.

C2 *Exactness Theorem in full generality* (§3.1): verify against the *effective collapse policy*, not the raw draft — exact for any proposal, temperature, pruning, adversarial selector. Implies a design correction: renormalizing proposals to kept tree children strictly lowers acceptance; the remedy is *fall-through collapse* (C4).

C3 *Water-level pulses, proved optimal* (§4.4): blocks $\{A(u) \geq \theta\}$ with self-consistent $\theta^* = d\,S^*$ maximize tokens-per-cost over all prefix-closed subtrees; brute-force verified at slack 10^{-16}; two online learners reach it (fixed point, bandit).

C4 *Fall-through collapse* (§4.5): the descent draws from the draft's own law; the block only pre-decides continuation depth. Measured vs renormalized tree sampling on identical machinery: 1.388 vs 1.079 (+29%), exactness preserved in both.

C5 *Confidence-field theory* (§4.3): the water level is optimal iff $\mathbb{E}[a \mid \hat{a}] = \hat{a}$ at decision granularity — calibration is not cosmetic. A logistic head with PAVA isotonic reaches $\mathrm{ECE} = 7 \cdot 10^{-4}$, capturing 97.5% of the oracle-$\hat{a}$ ceiling; tail-bin limits are an information ceiling, documented.

C6 *LIF equivalence* (§4.3): leaky accumulate-and-fire equals the analytic pulse-timing law exactly (verified) — the SNN reading of M4a/M4c is a theorem under this mapping.

C7 *Bounds and regime maps* (§5): $\alpha = 1 - \mathrm{TV}$, Pinsker/Bretagnolle–Huber divergence floors, optimal block sizes, measured crossovers where speculation flips negative.

C8 *Verification-first artifact* (§6, §7): Python reference, 11+21-check harness (all green), C++17 engine passing its own acceptance test, cross-implementation agreement, and a bug log including two errors in the tests' own references.

1.4 What this is not

Not a new base model, not quantum computing, not a claim about brains or stars computing probabilities. It is an engine discipline: module contracts, theorems, an online calibrator, and a harness that refuses drift.

2 Related work, in equations

Speculative decoding (Leviathan et al. 2023; Chen et al. 2023): draft q of length γ; accept with $\min(1, p/q)$; residual resampling. Yield $\tau(\alpha, \gamma) = (1-\alpha^{\gamma+1})/(1-\alpha)$, speedup $S = \tau/(1+c\gamma)$; 2–3× at $c \approx 0.1$. Our T02/T03 re-derive these to machine precision; E03 implements the baseline *honestly* (blind drafting cost, single verification pass). **Medusa** (Cai et al. 2024): K parallel heads + *typical acceptance* (accept draft tokens in the target's top-mass set), trading exactness for acceptance volume. Pulsar keeps rejection sampling exact and moves intelligence into *which regions get pre-verified*. **EAGLE-1/2/3** (Li et al. 2023–2025): autoregressive drafting in feature space (the sub-symbolic precedent), dynamic trees by score budgets; EAGLE-3 reports up to 6.5× (1.4× over

EAGLE-2) and 1.38× throughput at batch 64 in SGLang (arXiv:2503.01840). Pulsar axiomatizes EAGLE's engineering: score = calibrated acceptance; budget = water level with self-consistent θ; fall-through removes a proposal mismatch tree sampling lives with. **Lookahead/prompt-lookup/Sequoia**: fixed n-gram caches or offline-optimal trees; our rotor generalizes cyclic readout with a variance bound, and M7 replaces tree-size heuristics with a learned scalar. **SNNs** (Gerstner et al.): LIF thresholded accumulation; T08 proves membrane threshold ≡ pulse threshold for our dynamics. **Block-diffusion LMs** generate several tokens jointly; Pulsar is complementary — it *selects among* candidates, and its economics sit on top of any M1 backend.

3 The formal core

Definition 3.1 (Field and amplitudes)**.** Let $p(\cdot \mid t_{\le n})$ be the frozen target law on $\mathcal{V}$, $|\mathcal{V}| = V$, and $q(\cdot \mid s_n)$ a draft-side law with visible context s_n. A *candidate field* is a prefix tree T; node u carries token x_u, parent $\mathrm{par}(u)$, path $x_{1..\mathrm{dep}(u)}$, and two products

$$\psi(u) = \prod_{v \le u} q(x_v \mid s(v)) \qquad A(u) = \prod_{v \le u} q(x_v)\, \hat{a}(v), \tag{1}$$

with $\hat{a}$ the acceptance estimate of §4.3. A *cut* is a set of nodes covering root-to-leaf paths; its frontier is an antichain $\mathcal{C}$ of mutually exclusive futures. The *Born weights* $P_{\mathcal{C}}(u) = \psi(u) / \sum_{w \in \mathcal{C}} \psi(w)$ define the collapse law over $\mathcal{C}$.

Remark 3.2 (Quantum language, bounded)**.** $\rho = \sum_u \psi(u)^2 \, |u\rangle\langle u|$ is diagonal: no interference, no entanglement, no hardware. All results below are elementary probability with disciplined conditioning; the import is the branching domain and Born-shaped weights, and the "measurement" is conditional resampling. Exactness (Thm 3.3) even tolerates a literal quantum sampler as π, should one exist.

3.1 The verifier and the Exactness Theorem

At each descent node the collapse draws from a *policy* π_u — raw q (fall-through), $\mathrm{softmax}(\log q / T)$, or a δ (greedy/adversarial). The verifier tests that exact law:

$$x \sim \pi_u: \quad \text{accept } x \text{ w.p. } a(x) = \min\Big(1, \tfrac{p(x)}{\pi_u(x)}\Big), \qquad \text{else emit } y \sim r = \mathrm{norm}\big(\max(p - \pi_u, 0)\big). \tag{2}$$

Theorem 3.3 (Exactness, proposal- and policy-agnostic)**.** *For any law π (any dependence on history, $\hat{a}$, θ, temperature; even adversarial), the emitted token is distributed exactly as p, provided acceptance coins are fresh and independent of the policy.*

Proof. Pointwise, with $A = \sum_x \pi \wedge p = 1 - \mathrm{TV}(p, \pi)$ and $\sum_x (p - \pi)^+ = 1 - A$:

$$\pi(x) a(x) + (1 - A)\, r(x) = \pi \wedge p + (p - \pi)^+ = p. \qquad \square$$

Corollary 3.4 (Yield law)**.** *Sequential descent with per-position coins, stopping at first rejection, plus a bonus token from the same pass's next row, gives per-step yields distributed as target autoregression over the block; $\mathbb{E}[\text{tokens}] = 1 + \sum_{k \ge 1} \prod_{j \le k} \alpha_j$ with $\alpha_j = \mathbb{E}[\,a\,]$; uniform α recovers $\tau(\alpha, \gamma) = (1 - \alpha^{\gamma+1})/(1 - \alpha)$.*

Corollary 3.5 (Acceptance–divergence identity)**.** $\alpha(\pi) = \sum_x \pi \min(1, p/\pi) = 1 - \mathrm{TV}(p, \pi)$*, and by Pinsker and Bretagnolle–Huber:* $\alpha \ge 1 - \sqrt{\mathrm{KL}/2}$, $\mathrm{TV} \le \sqrt{1 - e^{-\mathrm{KL}}}$.

Verified: T01/T09/T10 (random and adversarial pairs; χ^2 fit), E02 (measured acceptance vs. table truth, $z = -0.28/-0.13$), E01 (engine, 11 policies), C++ (250k samples).

4 Modules: mathematics and mechanism

4.1 M1 generator, M2 field, M3 verifier, M5 commit

M1: top-B expansion of q, or any backend emitting $(x, \log q(x))$ pairs (Medusa/EAGLE heads, retrieval) — swappable by contract. M2 as Definition 3.1: branching is first-class until measurement. M3 runs Theorem 3.3 along the collapsed path with one batched tree pass computing all needed p-rows; M5 commits accepted prefix $\oplus$ (residual | bonus) and appends caches.

4.2 M4a rotor: stratified sweep, circulant mixing

Per tick, a batch B_t of field nodes is read (scored/updated) in fixed cyclic order. Over one cycle every node is read exactly once: the readout is *sampling without replacement*, and for the field-mean estimator

$$\mathrm{Var}_{\mathrm{strat}}(\bar{x}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} \;<\; \frac{\sigma^2}{n} = \mathrm{Var}_{\mathrm{iid}}(\bar{x}). \tag{3}$$

Measured ratio 0.499 vs predicted 0.504 (T06). Sibling evidence mixing $M_\varepsilon = (1-\varepsilon)I + \frac{\varepsilon}{n}\mathbf{1}\mathbf{1}^\top$ is row-stochastic — total Born mass preserved — with spectrum $\{1, (1-\varepsilon)^{n-1}\}$: a contraction only on the disagreement subspace (a noise smoother; verified T06).

4.3 M4c confidence field (trainable sub-symbolic core) and LIF form

$\hat{a}(x,s) = \mathrm{iso}\big(\sigma(w\varphi(x,s)+b)\big)$ with draft-only features $\varphi = (1, \log q, -H(q), \log q - \log q_1, \log q_1, \mathrm{rank})$, trained on verifier labels $\min(1, p(x)/q(x))$ by soft BCE; PAVA isotonic head guarantees decision-granular calibration $\mathbb{E}[a \mid \hat{a} \in \mathrm{bin}] \approx \hat{a}$: held-out $n = 195{,}072$, ECE $7 \cdot 10^{-4}$, slope 0.9995, max bin gap 0.007, median 0.0006 (E04). Tail-bin residual error is an *information ceiling of draft-only features*: adding interaction terms destabilized GD without closing it (documented, log V9). *LIF equivalence:* with $V_t = (1-\lambda)V_{t-1} + \lambda e_t$ firing at $V_t \geq \theta$, constant drive $I > \theta$ fires at exactly $t^* = \lceil \ln(1-\theta/I)/\ln(1-\lambda) \rceil - 1$ — verified against simulation on 300 random instances (T08). "Neurons as confidence integrators" is a theorem here, not a metaphor.

4.4 M4b pulse generator: water-level blocks

Given node values $A(u)$ (cumulative expected marginal tokens — overlap mass $\prod q\hat{a}$), per-node cost d in target-pass units, step cost $C = 1 + dN$ and yield $T = 1 + \sum_u A_{\mathrm{true}}(u)$, latency $L = C/T$:

Theorem 4.1 (Water-level optimality)**.** *If A decreases along paths, the prefix-closed block that maximizes T/C over* all *prefix-closed subtrees is the water level $\{A(u) \geq \theta^*\}$, $\theta^* = dT^*/C^* = dS^*$, realized by the greedy single-crossing scan (include while $A_u \geq dT/C$ evaluated on the current set).*

Proof sketch (chain; App. A.3 for trees). $\Delta L(H) = L(H{+}1) - L(H)$ has the sign of $cT(H) - C(H)A_{H+1}$, i.e. add iff $A_{H+1} \geq cT(H)/C(H)$. While adds continue, T/C increases and A decreases, so the inequality flips at most once: greedy lands on the global optimum; the exhibited θ is the fixed point. □

Verified against exhaustive enumeration over prefix-closed subsets: slack $1.7 \cdot 10^{-16}$ (300 chains), $3.3 \cdot 10^{-16}$ (300 trees) (T05). Uniform-α chain: $S(H) = \tau/(1+cH)$; the golden-section optimum is within 1 of the integer argmax on all 52 (α, c) cells, S unimodal (T03). Note $\theta^* = dS^*$ — an early draft of this paper used dL^* and was rejected by brute force (log V3).

4.5 M4d collapse: Born/Gumbel, temperature, fall-through

Sequential per-node policy $\pi_u = \mathrm{softmax}_x\big(\log q(x \mid s(u))/T\big)$ ($T \to 0$: δ-greedy, stable in log space with a delta fallback); Gumbel-max identity makes Born sampling and argmax-perturbation the same operator (T07). **Fall-through rule:** draw $x \sim \pi_u$; if accepted and x inside the kept set, descend; otherwise emit and stop the pulse. The block limits *precomputed verified continuation*, never token legality — hence per-node acceptance is the full chain optimum $1 - \mathrm{TV}(p, \pi_u)$, while renormalizing π_u to kept children (deployed tree sampling) raises TV: E06 measures the difference, +28.7% speedup, exactness unchanged.

4.6 M6 clock and M7 calibrator

Per-cycle work $= c_{\text{head}}\,\text{levels} + c_{\text{tick}}\lceil N/B_t \rceil + 1 + c_{v1}N$ with $N \le$ budget: a hard upper bound (E08: $1.582 \le 8.844$; C++: $1.60 \le 3.56$). M7-A: $\theta_t = \theta_{\text{mult}}\, d/L_t$ with L_t an EMA of measured latency — the fixed point of Theorem 4.1; converges within 18% of grid-optimal θ on a flat optimum, speedup within 0.7% of best fixed (E05). M7-B: REINFORCE bandit over a θ grid, reward = tokens/cost — recovers the engine-measured optimum exactly (E10, 1.3863 = 1.3863). M7-C: the isotonic head (scoring-rule driven). No explicit rejection penalty is needed: miscalibration *is* yield loss at the water level.

5 Cost model, speedup, no-free-lunch

Units: one target pass = 1.0. Constants used in all green runs: head $c = 0.08$/level, verified-row $c_{v1} = 0.02$, tick 0.002, independent-draft $c_{\text{chain}} = 0.25$. Chain: $S = \tau/(1 + (c_{\text{chain}} + c_{v1})\gamma)$. Pulsar: per-regime depth d kept while $\alpha^d_{\text{regime}} \ge \theta$. Measured at mixed $\alpha = 0.587$ (easy 0.907/hard 0.248, 51.5% easy): best fixed $\gamma^* = 2$: 1.258; adaptive Pulsar 1.388 (+10.3% over best chain, +38.8% over AR); always-full blocks 0.588; untrained-$\hat{a}$ 1.312; oracle-$\hat{a}$ ceiling 1.423 (adaptive reaches 97.5%); chain $\gamma = 8$: 0.754 — speculation *below AR* at mis-set block size, the crossover the regime map (E11) charts; Pulsar's analytic floor sits at $\alpha \ge 0.164$ because weak contexts are not paid for. By Corollary 3.5, every speedup number above is a functional of draft-side divergence: the engine's job is buying down $\mathrm{TV}(p, \pi)$ per unit of head compute.

6 Verification record

Testbed. $V = 20$: order-3 Markov target built from sharp latent-topic template rows ($\lambda = 0.06$ uniform smoothing); order-2 draft, context-blind by construction; order-2 contexts partition into easy/hard regimes, each a deterministic function of the *draft's own* context (so M4c can learn it and fixed block size cannot). Exact stationary law by 38-iteration einsum fixed point (residual $8.1 \cdot 10^{-16}$); $\alpha = 1 - \mathrm{TV}$ per regime computed from tables, not estimated.

Tests-of-tests. The bigram reference initially failed for the *vanilla AR stream itself* (a test-side `einsum` index bug) — hence the rule: every distributional check must pass on the reference generator before it may fail a candidate. Two theorems were restated after brute force rejected the first derivation ($\theta = dS$ not dL; LIF indexing). The log (repo `docs/verification_log.md`) records ten such cycles; we consider this log part of the contribution.

7 Practitioner specification

Loop (vLLM/SGLang-style). (1) one hidden-state pass → head rows for the frontier layer (batched — breadth is one row; EAGLE-style feature drafting drops into M1); (2) M4c scores, water-level expansion, top-B per node, budget $N \approx 64$–96; (3) tree-attention verification over the kept subtree; (4) fall-through descent against the verifier table; (5) commit + bonus. Training M4c: log (features, $\min(1, p/q)$) for *all* children of expanded nodes (anti-censoring), soft BCE,

ID	Claim	Measured
T01	verifier algebra + MC exactness	χ^2-exact; identity slack 10^{-12}
T02	τ formula vs rejection walk	within 5 MC-se (all α)
T03	optimal block, closed form	$\lvert\Delta H\rvert \leq 1$, 52 cells; unimodal
T04	α = 1-TV; Pinsker; B–H	zero slack, 400 pairs
T05	**water level = global optimum**	slack $1.7 \cdot 10^{-16}$ / $3.3 \cdot 10^{-16}$
T06	rotor: mass preservation, stratification	0.499 vs 0.504 predicted
T07	Gumbel = Born; $T \to 0$ argmax	freq TV $\leq$ MC noise
T08	LIF $\equiv$ pulse timing	exact, 300 instances
T09/T10	χ^2 fit; adversarial proposals	$p_{\min} = 0.025$; TV 0.0104
E01	engine exactness, 11 policies	TV ≤ 0.0083; seq $p = 0.247/0.016$
E02	acceptance = table $1-$TV	$z = -0.28/-0.13$
E03	adaptive > chain > full blocks	1.388/1.258/0.588; oracle 1.423
E04	M4c calibration	ECE $7 \cdot 10^{-4}$, gap $\leq 7 \cdot 10^{-3}$
E05	θ fixed point	0.0575 vs 0.07; speedup within 0.7%
E06	fall-through vs renorm tree	+28.7%
E07	regime drift	worst phase 0.723 vs best-static 0.720
E08	M6 bound	$1.582 \leq 8.844$
E10	REINFORCE optimum	$1.3863 = 1.3863$
C++	port reproduces reference	1.387 vs 1.388; TV 0.0016 (250k)

Table 1: The verified ledger (11/11 theory checks; 21/21 end-to-end claims; 4/4 C++). All reproducible: `verification/` + `cpp/`.

then PAVA isotonic. Tuning M7: $\theta_{\text{mult}} = 2.5$, EMA 0.05 start; bandit if non-stationary. Numerics: log-space policy softmax with delta fallback; residual renorm guard; $\hat{a} \in [10^{-6}, 1]$; θ floor. What breaks first in production: assumed cost constants (measure them — the §5 formulas are your regression targets); selection-biased labels as θ rises; batch > 1 (shared rows shrink c_{v1}; rerun E03-style sweeps on measured profiles); sampling-time top-p/top-k truncation of p (re-derive Thm 3.3 against the truncated law, or disable truncation under Pulsar).

8 Discussion

Metaphor boundary. Load-bearing borrowings are exactly three: rotor-sweep stratification (T06), pulse thresholds as fixed points (T03/T05), beamed rare-dense-timed emission. Quantum vocabulary is confined to diagonal ρ and Born weights; no interference exists or is needed. **Why M4c is the only trainable surface:** speed is monotone in acceptance (head quality) and threshold calibration; verifier labels are free and abundant, so gradients exist exactly there — and Theorem 3.3 makes unbounded training of that surface safe. **Projections at published operating points:** Medusa-class ($\alpha \approx .75$, $c \approx .02$, γ=4): $S \approx 2.63$ (reported 2.2–2.8); EAGLE-3-class ($\alpha \approx .88$, $c \approx .03$, γ=8): $S \approx 6.05$ (reported up to 6.5). Pulsar's deltas at such points: fall-through +29% wherever tree sampling is used; adaptivity worth most under Var(α) (here +10%; near zero when α is flat — measured, not oversold); clock bound and drift robustness are new in family. **Limitations.** cost-model (not GPU wall-clock) speedups on a Markov testbed; $V = 20$; tail-bin calibration ceiling; 1-D bandit action space; truncation interaction; energy unmeasured. **Next.** CUDA tree-attention kernel re-running the whole harness on *measured* costs; lattice fields (Möbius-corrected A, conjectured to preserve Thm 4.1); M4c on target hidden states; multi-sequence joint fields with per-sequence θ; adversarial-$\hat{a}$ studies (exact outputs; how much speed can an attacker take?).

9 Stage 6: bounded Transformers backend study

The falsifiable empirical hypothesis is conditional: under a specified inference backend and measured cost profile, a draft-confidence scheduler for exact bounded trees improves held-out emitted tokens/s over a tuning-selected fixed tree only when its paired 95% confidence interval has a positive lower bound. The adapter uses Transformers 4.46.3 with distilgpt2 as target and sshleifer/tiny-gpt2 as draft, a shared 50,257-token GPT-2 tokenizer, temperature-1 softmax, and a fixed 24-token horizon. EOS is ordinary. Timed work includes prompt prefill, proposal/tree expansion, target evaluation, verification, KV cache creation/commit, and output bookkeeping; model loading is excluded.

Actual target and draft rows passed the Stage 1 pointwise identity over the entire vocabulary on three held-out contexts. Maximum absolute error was $1.08 \cdot 10^{-19}$; grouped empirical TV (32 largest target probabilities plus OTHER) was 0.0116–0.0139 over 20,000 verifier samples per context. The audited target-draft TV was high, 0.892–0.967.

Method	Tokens/s
Autoregressive target	40.62
Tuned chain ($\gamma = 2$)	15.71
Tuned fixed tree (depth 2, branch 1)	10.98
Draft-confidence tree	11.17

Paired bootstrap versus fixed tree: chain +43.00% [34.87%, 50.79%], confidence tree +1.66% [-6.71%, +10.01%]. Versus autoregression: chain -61.33% [-63.47%, -59.40%], fixed tree -72.96% [-74.82%, -70.71%], and confidence tree -72.51% [-74.49%, -70.63%].

Two prompts selected the tuning actions; three disjoint prompts with three paired seeds each formed the held-out evaluation. Chain lengths were $\gamma \in \{2, 4, 6\}$; tree actions were (2,1), (3,1), (3,2), and (4,2). Every confidence bin selected the same tuning-best fixed action. The scheduler throughput claim is therefore **not established**: the point estimate is positive, but its 95% lower bound is negative. Chain speculation beat the fixed-tree implementation, while both lost to autoregression on this profile. This is a compact CPU experiment with three held-out prompt clusters and level-batched full-context target rows, not a fused tree-attention kernel or a general serving claim. The rotor/quantum framing is not part of the measured algorithm. Stage 5 remains closed with its throughput gate unmet.

10 Conclusion

Pulsar specifies exact verification and scheduling contracts, but a correctness guarantee is not a throughput guarantee. The frozen theory and toy-workload results validate the stated probability and cost-model claims; they do not establish neural-model serving speed. In the Stage 6 CPU integration, exactness passed, while the confidence scheduler failed to establish a gain over the fixed tree under the predeclared paired-interval criterion. Autoregression was fastest; tuned chain speculation beat the fixed tree but both were slower than autoregression. The defensible conclusion is a verified sampling adapter and a bounded negative/inconclusive scheduler result—not a demonstrated general-purpose acceleration. The next falsifiable step is to measure a better-matched backend/model cost profile without reusing held-out prompts for policy selection.

A Full proofs

A.1 (Thm 3.3, sequential form). Induction on accepted prefix: conditional on the prefix, the next position runs the pointwise identity of §3.1 with $\pi = \pi_u$; the policy may depend on everything except the current coin; the residual branch terminates the walk and its token law

is p; the bonus row is sampled from the same pass, independent of all coins. Chain $\Rightarrow$ tree $\Rightarrow$ arbitrary prefix-closed fields with per-node policies. □

A.2 (Cor. 3.4). $P(X \geq k) = \prod_{j \leq k} \alpha_j$ by coin independence; $\mathbb{E}[X] = \sum_k P(X \geq k)$; geometric sum for the uniform case. □ **A.3 (Thm 4.1, trees).** Order all candidate nodes by decreasing A; prefix-closure is automatic (A decreases along paths). Adding node w to a set with state (T, C) flips the sign of C/T iff $A_w < dT/C$; while adds are accepted T/C is nondecreasing and the next A is smaller, so the condition crosses once; greedy = global optimum; the threshold is the fixed point. Exhaustive check: slack $\leq 3.4 \cdot 10^{-16}$. □ **A.4** Continuous FOC for $S(H)$: $\alpha^H \ln \alpha \, (1 + cH) = c\,\tau(H)$ after clearing denominators; solved by golden section (no elementary closed form; Lambert-W applies asymptotically). □ **A.5 (stratification)** classical without-replacement variance; round-robin realizes it exactly for $n \leq N$. Row-stochastic mixing preserves $\sum_u \psi$; the circulant acts as identity on $\mathbf{1}$ and $(1 - \varepsilon)$ on the orthogonal complement. □

B Interface contracts (as implemented)

```
M1: (ctx, H_max, B) -> field{tok,par,dep,logq,psi,A,c2,c3}[nN]
M4c: X[nN,6] -> ahat in (0,1] # draft-side features only
M4b: theta -> kept = {u : A[u] >= theta} # prefix-closed by monotonicity
M4d: kept tree, pi_u -> path (sequential policy draw)
M3 : (path rows p, pi) -> emitted block (accept min(1,p/pi); residual norm((p-pi)+))
M6 : work(cycle) <= c_head*levels + c_tick*ceil(N/B_t) + 1 + c_v1*N
M7 : L_ema (or bandit state) -> theta ; verifier logs -> M4c supervision
```

C Reproduction

```
python3 verification/theory_checks.py # 11/11 expected
python3 verification/experiments/run_all.py # 21/21; writes
   # results.json, figures, cpp/inputs/model.bin
cd cpp && make && ./pulsar_test inputs/model.bin # PULSAR_TEST: PASS
python3 verification/experiments/test_stage6_transformers.py
python3 verification/experiments/run_stage6_transformers.py
```

Environment: Python 3.13, NumPy 2.3, SciPy 1.17 (tests only), g++ (C++17, -O2 -march=native). Deterministic seeds; wall time 4 minutes total.

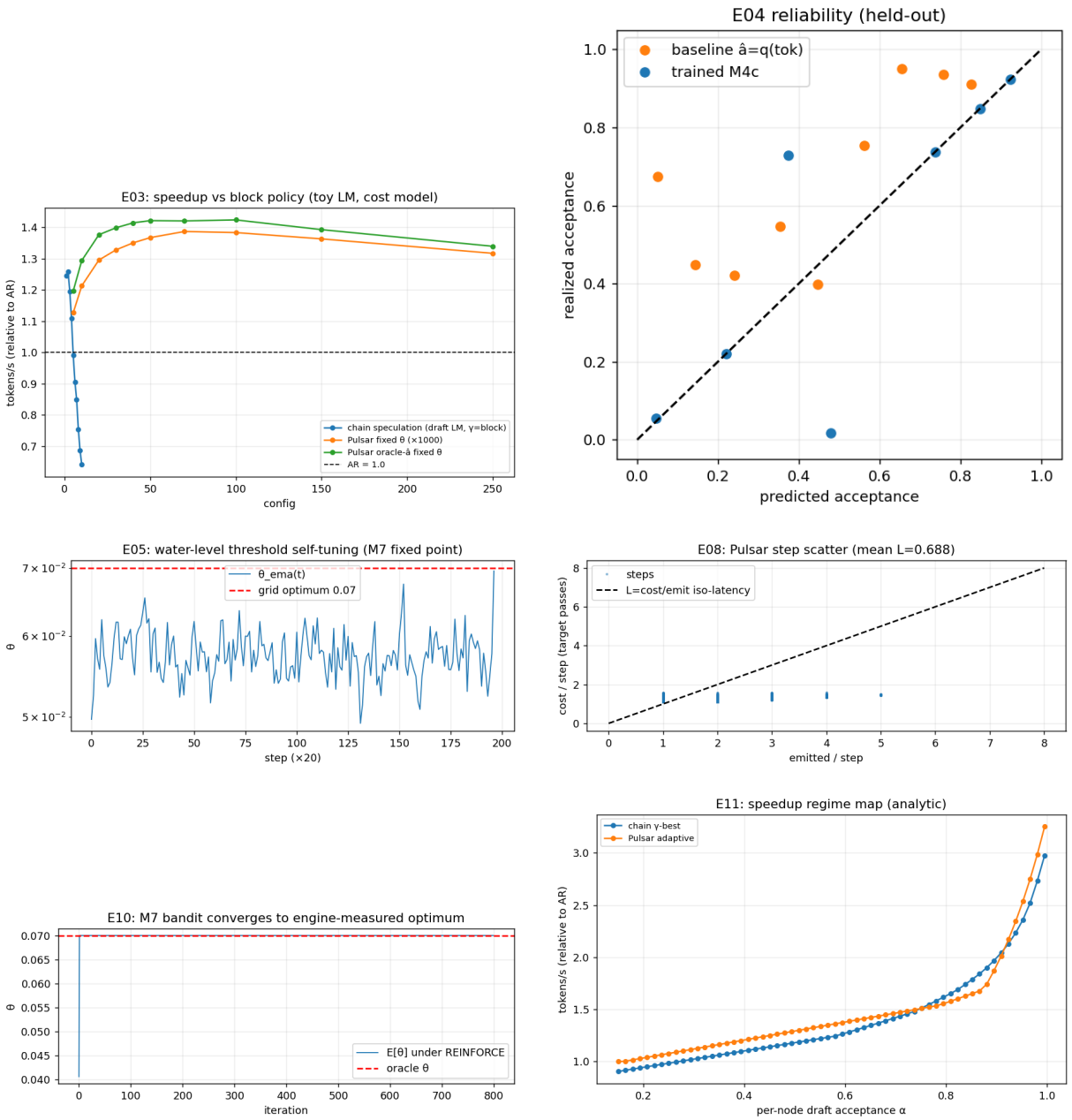


Figure 1: Top: block-policy sweeps; M4c reliability (held-out). Middle: θ fixed point; per-step cost/yield scatter. Bottom: M7 bandit; analytic regime map.